Multiobjective Optimization of Hydrogen Injection into Natural Gas Pipelines

A Surrogate-Assisted Approach

CLD880 - Minor Project Presentation

Sakhare Yash Balram (2022CH71496)

Supervisor: **Prof. Abhijeet Raj**

Department of Chemical Engineering
Indian Institute of Technology Delhi

May 2026

The Problem: H_2 Blending into Gas Grids

Hydrogen-Natural Gas Blending (HBNG) is the cheapest path to decarbonise the gas grid without rebuilding it.

The density mismatch drives stratification:

- H_2 is **8×** **lighter** than CH_4 (0.082 vs 0.668 kg/m^3).
- Buoyancy pushes H_2 to the pipe crown instead of mixing uniformly.

Property	CH_4	H_2	Ratio
Density (kg/m^3)	0.668	0.082	8×
Vol. energy (MJ/m^3)	36	11	3×
Flammability (%vol)	5-15	4-76	5× wider

Downstream Consequences

- **Hydrogen embrittlement** at the pipe crown (localised high p_{H_2}).
- **Wobbe Index drift** → combustion instability at end-burners.
- **Metering errors** from composition-dependent speed of sound.

Uniform mixing is a safety and regulatory requirement, not just an optimisation goal.

Our Contribution - Closing the Gap

Gap: No study treats injection angle α as a primary design variable, and none performs bi-objective (CoV, ΔP) optimisation.

This project:

1. Sweep $(\alpha, d/D, \mathrm{VR})$ on equal footing.
2. Top vs. bottom injection (buoyancy control).
3. 4 campaigns, 40 validated CFD cases.
4. GPR surrogate + NSGA-II Pareto search.

Design Variables:

Variable	Symbol	Range
Injection angle	α	30°, 90°
Diameter ratio	d/D	0.15-0.45
Velocity ratio	VR	0.69-5.84
Injection side	-	Top / Bottom

Formal MOO:

$$\min_{\mathbf{x}} \ \mathbf{F}(\mathbf{x}) = \begin{bmatrix} \mathrm{CoV}(\mathbf{x}) \\ |\Delta P|(\mathbf{x}) \end{bmatrix}$$

First simultaneous optimisation of mixing quality and pumping cost for T-junction H_2 injection.

Methodology Overview

Solver & Physics:

- OpenFOAM `rhoReactingBuoyantFoam`
- k-ω SST, compressible, buoyancy-coupled
- Half-domain symmetry ($x \geq 0$)

Operating Conditions:

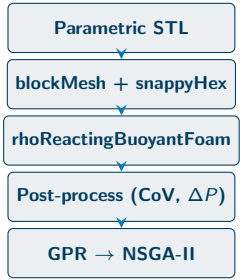
- Main: $D = 0.46$ m, $u = 10$ m/s, CH_4, $P = 6.9$ MPa
- Branch: H_2, $u_{br} = \mathrm{VR} \cdot u_{main}$

Metrics:

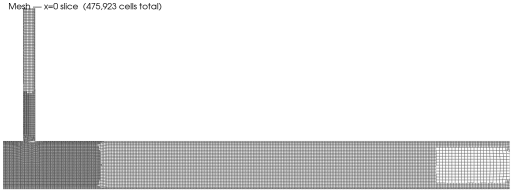
- **CoV** $= \sigma(Y_{H_2})/\bar{Y}_{H_2}$ at outlet
- $|\Delta P|$ = LP-filtered pressure drop on $[0.4, 1.2]$ s

Design of Experiments:

- Sliced LHS: 5 d/D bins $\times$ 2 VR per bin = 10 cases/campaign
- Same seed $\rightarrow$ matched-pair control
- 4 campaigns: $\{30^\circ, 90^\circ\} \times \{\text{top, bottom}\}$



Mesh and Validation



Mesh on the $x = 0$ symmetry plane (90°, case 03). Boundary layers visible at walls.

Mesh statistics (typical):

Half-domain cells	$\sim$463 k
Max non-orthogonality	49°
Layer coverage	99.6%
y^+	$\sim$1

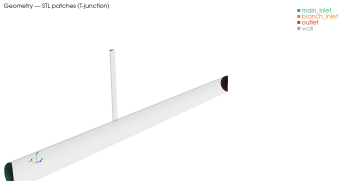
Grid independence (3 levels):

Mesh	ΔCoV	$\Delta\lvert\Delta P\rvert$
230k $\rightarrow$ 460k	0.4%	1.6%
460k $\rightarrow$ 920k	0.2%	0.7%

All 40 cases pass:

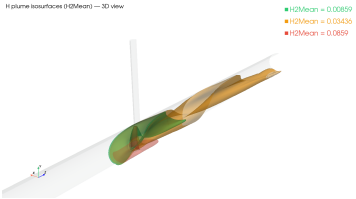
- Cumulative continuity error $< 10^{-2}$
- Residuals $< 10^{-8}$
- Mass balance $< 0.5\%$
- No NaN / FPE

Results Walkthrough: Case 04 (90° Top, VR = 3.81, CoV = 0.066)

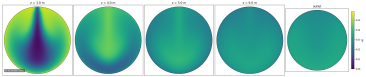


(a) T-junction geometry

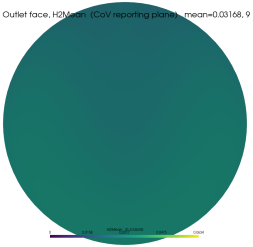
(b) Mesh on $x = 0$ plane (476k cells)



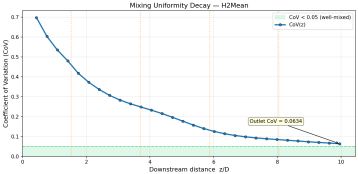
(c) H_2 isosurfaces (3D plume)



(d) H_2 at $z = 3, 4, 5, 6$ m + outlet



(e) Outlet: nearly uniform

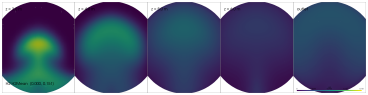


(f) CoV: 0.70 → 0.063

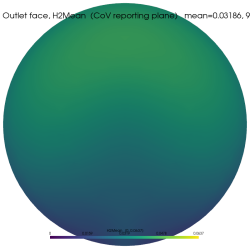
Same Case, Bottom Injection: CoV = 0.066 → 0.271 (4× worse)

Same 90°, $d/D = 0.252$, VR = 3.81 - only injection side changed.

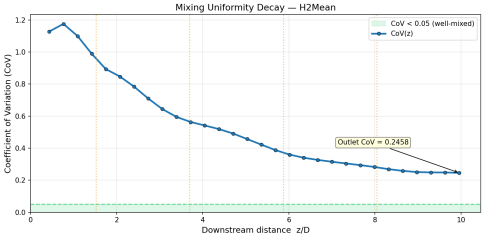
(a) H_2 plume rises as narrow column



(b) Crown streak persists to outlet



(c) Outlet: strong gradient remains

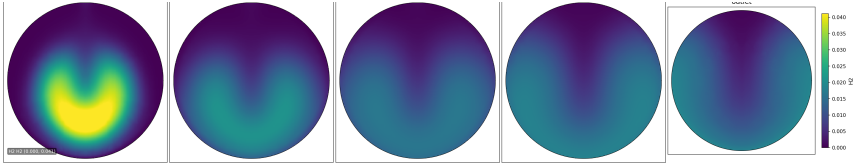


(d) CoV → 0.246 - never reaches 0.05 target

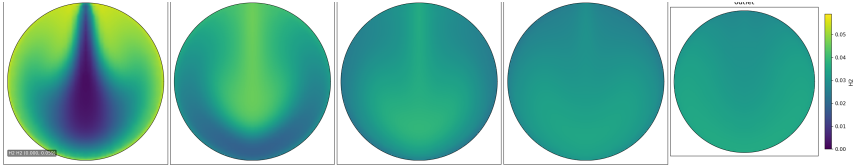
All 17 matched pairs: top injection wins every time ($p = 1.5 \times 10^{-5}$). Median degradation: +109% at 90°, +324% at 30°. $|\Delta P|$ is unchanged.

Velocity Ratio Effect: Same Geometry, Different VR

90° top, $d/D = 0.252$ - only VR differs. Cross-sections at $z = 3, 4, 5, 6$ m and outlet.



Case 03: **VR = 1.33**, CoV = **0.472** – H_2 hugs the crown; dark bottom = unmixed CH_4.

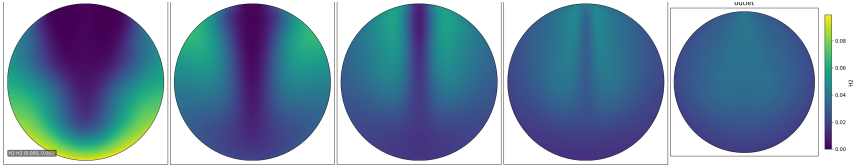


Case 04: **VR = 3.81**, CoV = **0.066** – Jet penetrates past centreline; uniform by $z = 5$ m.

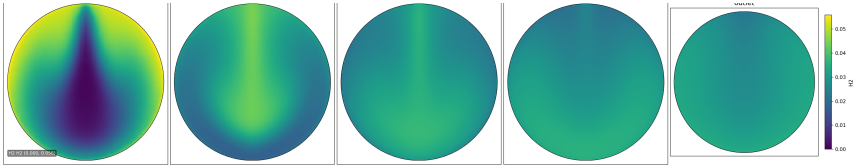
Same pipe, same branch. Tripling VR (1.33 → 3.81) cuts CoV by 7×. VR is the strongest lever on mixing.

Angle Effect: Same Design Point, 90° vs 30°

$d/D = 0.296$, VR $= 2.61$, HBR $= 0.187$ - only angle differs. Cross-sections at $z = 3, 4, 5, 6$ m and outlet.



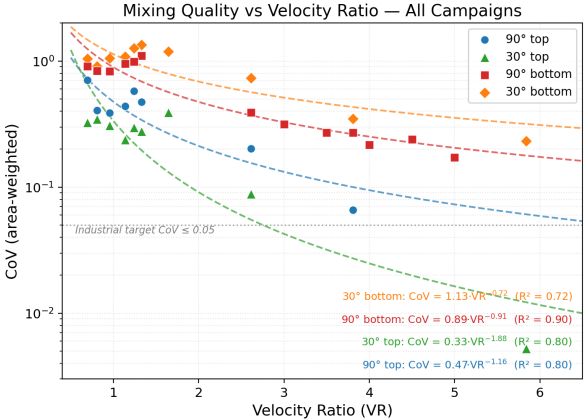
90°, case 06: CoV = **0.202** – Vertical streaks persist to outlet; crown-bottom asymmetry remains.



30°, case 06: CoV = **0.088** – Teardrop plume at $z = 3$ m homogenises by $z = 5$ m; outlet is uniform.

Identical design point, only angle changed. All 7 matched pairs: median CoV drops 46%, $|\Delta P|$ drops 63% ($p = 0.008$).

Power-Law Scaling: CoV vs VR



Log-log fit of CoV vs VR for all four campaigns. Dashed line: CoV = 0.05 target.

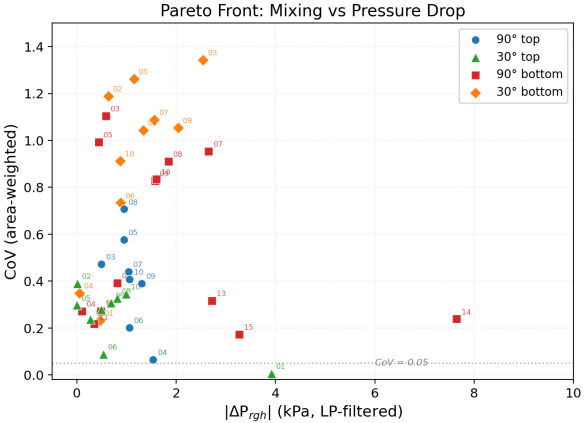
Campaign	Fit	R^2
90° top	$0.47\,\mathrm{VR}^{-1.16}$	0.80
30° top	$\mathbf{0.33}\,\mathrm{VR}^{-\mathbf{1.88}}$	**0.80**
90° bot	$0.89\,\mathrm{VR}^{-0.91}$	0.90
30° bot	$1.13\,\mathrm{VR}^{-0.72}$	0.72

The exponent for 30° top ($b = -1.88$) is 1.6× steeper than for 90° top ($b = -1.16$).

Each unit of VR buys **nearly twice as much mixing improvement** on the tilted geometry.

Bottom-injection exponents are shallower – buoyancy limits how much mixing VR can buy.

Pareto Front: The Trade-Off Visualised



Four-campaign Pareto front. Lower-left is better (low CoV, low $|\Delta P|$).

30° **top** occupies the desirable lower-left corner.

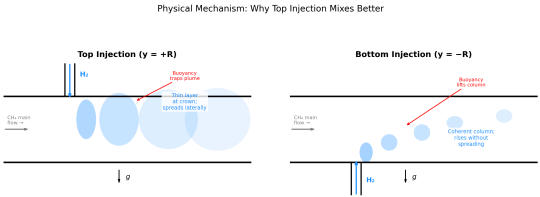
Standout cases:

- **Best mixing:** Case 01 (30° top) – CoV $= 0.005$ at $|\Delta P| \approx 4.2$ kPa.
- **Cheapest:** Case 09 (30° top) – $|\Delta P| \approx 0.3$ kPa with CoV $= 0.31$.

Every 90° point is dominated by some 30° point. No bottom-injection point reaches CoV ≤ 0.05.

30° top is the only configuration that consistently reaches CoV < 0.05.

Physical Mechanism: Why Does Tilt + Top Work?



Buoyancy mechanism. Left: top injection spreads H_2 laterally. Right: bottom injection lifts a coherent column.

Two synergistic effects:

1. Tilt (30° vs 90°):

- 90° jet impinges on opposite wall, traps a low-H_2 core.
- 30° jet feeds a streamwise shear layer entraining CH_4 along the pipe.
- Momentum aligned with flow $\rightarrow$ lower $|\Delta P|$.

2. Buoyancy (top vs bottom):

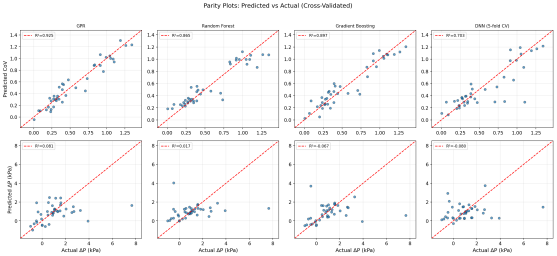
- Top: buoyancy *opposes* jet $\rightarrow$ thin wide layer, large diffusion area.
- Bottom: buoyancy *assists* jet $\rightarrow$ narrow column, small diffusion area.

Surrogate Model: Enabling Rapid Optimisation

Four architectures benchmarked on 40 cases:

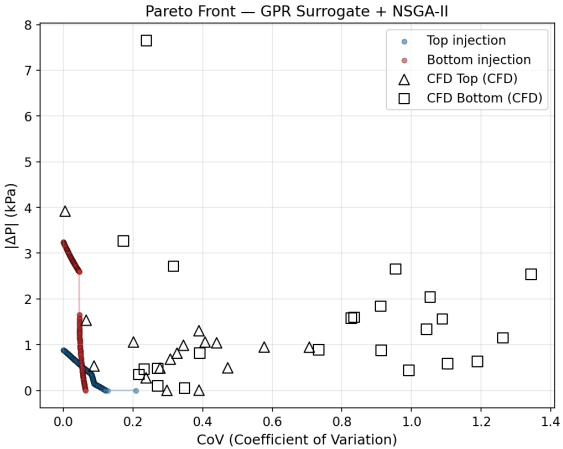
Model	Target	R^2	MAE
GPR	**CoV**	**0.925**	**0.083**
Grad. Boost	CoV	0.897	0.096
Random Forest	CoV	0.865	0.103
DNN (5-fold)	CoV	0.735	0.129
All models	$\lvert\Delta P\rvert$	$\approx$ 0.5-0.6	-

GPR (Matérn-5/2, LOO-CV) gives the best CoV prediction.
DNN ($\sim$17 400 parameters) overfits at $N = 40$.
$\lvert\Delta P\rvert$ prediction is weak - acoustic noise ($\pm$10-25 kPa) makes the $\sim$1-3 kPa signal hard to learn reliably.



Parity plots (predicted vs actual). GPR has the tightest scatter for CoV.

NSGA-II Optimisation: Pareto Fronts from GPR



NSGA-II Pareto fronts on GPR surrogate (200 pop, 150 gen). Hollow = CFD; solid = surrogate.

Knee-point designs (closest to utopia):

Top Injection Knee

$\alpha = 30^\circ$, $d/D = 0.45$, VR $= 5.82$
CoV ≈ 0.089, $|\Delta P| \approx 0.14$ kPa

Bottom Injection Knee

$\alpha = 90^\circ$, $d/D = 0.26$, VR $= 5.84$
CoV ≈ 0.048, $|\Delta P| \approx 0.97$ kPa

Both knee points maximise VR - consistent with SHAP: **injection side** and **VR** are the two dominant features.

Top injection achieves a better trade-off: lower $|\Delta P|$ for comparable CoV.

Limitations and Future Work

Current limitations:

- $|\Delta P|$ **surrogate unreliable** - acoustic noise in compressible transient solver.
- **Two angles only** (30°, 90°).
- **Single Reynolds number** ($u_{\text{main}} = 10$ m/s).
- $N = 40$ **cases** - insufficient for the DNN architecture ($\sim$17 k parameters).
- Mesh convergence verified at one design point; same baseline mesh used for all cases.

Planned next steps:

Full DoE (300 cases)
5 angles $\times$ 3 d/D $\times$ 2 sides
$\rightarrow$ train DNN to $R^2 > 0.95$

↓

Fix $|\Delta P|$
Steady RANS (`rhoSimpleFoam`)
or extend transient to $\sim$5 s

↓

Additional angles
45°, 60°, 120°, 150°
$\rightarrow$ full angular Pareto map

Conclusions

1. **First multi-objective CFD study** of H_2 T-junction injection with angle as a design variable. 40 validated RANS cases across 4 campaigns.
2. 30° **top injection dominates all other configurations.** Median CoV reduction: -46%; median $|\Delta P|$ reduction: -63% vs 90° (all 7 matched pairs).
3. **Top injection is strictly superior to bottom.** All 17 matched pairs favour top ($p = 1.5 \times 10^{-5}$). Median CoV degradation from top $\rightarrow$ bottom: $+109\%$ (90°), $+324\%$ (30°).
4. **Buoyancy is the key mechanism:** top injection spreads H_2 laterally (large interfacial area); bottom lifts a coherent column.
5. **GPR surrogate** achieves $R^2 = 0.93$ for CoV; NSGA-II Pareto search identifies optimal knee-point designs in seconds.
6. **All code, data and figures** are open-source at `https://github.com/Yash04dsr/MTP-Work`.

OpenFOAM Case Structure & Code Availability

Each DoE case directory:

```
case_XX/
 ├ 0/          Boundary conditions
 │  ├ U, p_rgh, H2, CH4, ...
 ├ constant/
 │  ├ polyMesh/  Mesh
 │  ├ thermophysicalProperties
 │  ├ turbulenceProperties
 ├ system/
 │  ├ controlDict, fvSchemes
 │  ├ fvSolution
 │  ├ snappyHexMeshDict
 ├ generateSTL.py
 ├ Allrun     Full pipeline script
 ├ case.env, case_info.json
 ├ figures/   13 PNG per case
 ├ postProcessing/
 ├ log.solver
```

DoE automation scripts:

- `lhs_design.py` - LHS sampling
- `stamp_cases.py` - case materialiser
- `run_doe.sh` - resilient orchestrator
- `all_metrics.py` - CoV, $|\Delta P|$
- `make_figures.py` - visualisations
- `cross_analysis.py` - campaign comparison
- `train_surrogate.py` - GPR + NSGA-II

All code, data, and figures:

`https://github.com/Yash04dsr/MTP-Work`

40 cases $\times$ 13 figures $=$ 520 visualisations,
plus cross-campaign analysis and surrogate model.

Boundary Conditions Summary

Velocity (U):

Patch	BC
main_inlet	$\frac{1}{7}$-power-law profile
branch_inlet	$\frac{1}{7}$-power-law profile
outlet	pressureInletOutletVelocity
wall	noSlip
sym	symmetryPlane

Species (H_2 / CH_4):

Patch	H_2	CH_4
main_inlet	0	1
branch_inlet	1	0
outlet	inletOutlet	inletOutlet
wall	zeroGradient	zeroGradient

Temperature: fixed at 288 K everywhere (energy equation disabled - isothermal).

Pressure (p_{rgh}):

Patch	BC
main_inlet	fixedFluxPressure
branch_inlet	fixedFluxPressure
outlet	prghPressure (69 bar ref)
wall	fixedFluxPressure

Turbulence (k, ω):

Patch	k	ω
Inlets	TI = 5%	$L_{\mathrm{mix}} = 0.07D$
outlet	inletOutlet	inletOutlet
wall	kqRWallFn	omegaWallFn

Per-case tokens (stamped by `stamp_cases.py`):

`@UMAIN@`, `@UBRANCH@`, `@D1@`, `@D2@`, `@ZJCT@`, `@K_MAIN@`, `@K_BRANCH@`, `@OMEGA_MAIN@`, `@OMEGA_BRANCH@`

Key References

- Y. Tian, T. Tian, G. Ren, and J. Zhang. "Numerical Simulation of Hydrogen Mixing Process in T-Junction Natural Gas Pipeline." *Energies*, vol. 16, no. 1, 2023.
- Y. Su, J. Li, W. Guo, Y. Zhao, et al. "Prediction of Mixing Uniformity of Hydrogen Injection in Natural Gas Pipeline Based on a Deep Learning Model." *Energies*, vol. 15, no. 22, 2022.
- A. K. Arya, R. Katiyar, P. Senthil Kumar, et al. "A multi-objective model for optimizing hydrogen injected-high pressure natural gas pipeline networks." *Int. Journal of Hydrogen Energy*, vol. 48, 2023.
- N. A. N. M. N. Azman et al. "Hydrogen-Natural Gas Co-Blending: Performance Analysis with CFD and Static Mixers." *Chemical Engineering Transactions*, vol. 122, 2025.

Thank You

Questions?

Sakhare Yash Balram

Department of Chemical Engineering
Indian Institute of Technology Delhi

ch7221496@iitd.ac.in

https://github.com/Yash04dsr/MTP-Work